

# Akshat Kumar Sinha

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## SUMMARY

Computer Science undergraduate specializing in Data Science with software engineering internship experience building enterprise applications using React, TypeScript, Python, and REST APIs. Hands-on experience in machine learning, natural language processing, computer vision, financial data analysis, and fraud detection. NPTEL-certified in Natural Language Processing, with experience training classification models, performing feature engineering, evaluating model performance, and working with SQL, PostgreSQL, MongoDB, OpenCV, Pandas, and Scikit-learn.

## EXPERIENCE

### Software Development Engineer Intern

Jun 2026 – Present

*Routa Digital India*

- Accelerated delivery of **6+ enterprise security modules** by developing reusable React and TypeScript components integrated with REST APIs, reducing average feature development time by approximately **30%**.
- Reduced server load by approximately **40%** on high-frequency reporting endpoints by implementing batch processing for bulk user operations and replacing per-record API calls with paginated and chunked request pipelines.
- Improved page load performance by approximately **35%** across three data-intensive dashboard modules by identifying excessive API requests and coordinating with backend engineers to consolidate REST operations.
- Implemented role-based access control across **6+ security modules** using permission-scoped components, dynamic user-group management, and role-conditional workflows for multi-level enterprise users.
- Collaborated with frontend and backend developers to debug API integration issues, optimize data flows, and deliver maintainable software features for a production cyber-risk management platform.

## PROJECTS

### Fraud Transaction Detection using Machine Learning

*Python, Scikit-learn, Pandas, NumPy*

- Trained supervised machine learning models to identify potentially fraudulent financial transactions using transaction-level data and engineered numerical features.
- Performed data preprocessing, missing-value handling, class-distribution analysis, feature selection, model training, and hyperparameter tuning.
- Evaluated classification models using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC to account for imbalanced fraud datasets.
- Authored an unpublished research paper documenting the problem statement, literature review, methodology, experimental setup, model evaluation, results, and limitations.

### Banking Market Intelligence Automation

*Python, Pandas, Financial APIs*

- Reduced manual financial data collection time by approximately **80%** by building an automated Python pipeline that retrieves, cleans, normalizes, and analyzes market data for **10+ banking stocks**.
- Generated structured Excel reports containing stock metrics, historical trends, comparative analysis, and summarized market insights.
- Applied Pandas-based data transformation, exception handling, time-series processing, and automated report generation to create a reusable analytical workflow.

### FI Calculations

*JavaScript, HTML, CSS, CSV Processing*

- Developed and deployed a financial calculator supporting **5+ investment metrics**, including SIP growth, compound interest, investment returns, loan amortization, and financial independence calculations.
- Designed modular calculation logic using object-oriented programming principles and implemented CSV-based data processing for structured financial input.

### Student Access Portal

*JavaScript, Node.js, JWT, RBAC*

- Built a three-role campus access platform for students, mentors, and administrators with JWT-based authentication and route-level authorization.
- Implemented role-specific dashboards and approval workflows to centralize temporary access requests, scheduling, and administrative resource management.

Self-Hosted Home Server

Ubuntu Server, CasaOS, Tailscale, Linux

- Configured and maintained a self-hosted Ubuntu Server with CasaOS and Tailscale, enabling secure remote access to locally hosted applications and services.
- Gained hands-on experience with Linux administration, networking, Docker-based services, system monitoring, service configuration, and troubleshooting.

RESEARCH EXPERIENCE

Anomaly Detection in Ethereum Transactions Using Autoencoder Networks 2026

- Developed an unsupervised deep-learning approach for detecting anomalous Ethereum accounts by training an Autoencoder exclusively on normal transaction behaviour and classifying anomalies using reconstruction error.
- Preprocessed a dataset of **9,841 Ethereum accounts**, reducing the feature set from 51 to **45 features** through missing-value handling, outlier clipping, normalization, and feature selection.
- Designed a **45–32–16–8–16–32–45** encoder-decoder architecture and optimized it using Mean Squared Error loss and the Adam optimizer.
- Achieved **69% precision**, **78.7% recall**, an **F1-score of 0.74**, and a **ROC-AUC of 0.70**, demonstrating the model's ability to detect suspicious transactions without requiring labelled data for training.
- Co-authored an academic research paper documenting the methodology, experimental results, limitations, and future improvements; the paper is currently unpublished.

EDUCATION

|             |                                                                                                                   |                        |
|-------------|-------------------------------------------------------------------------------------------------------------------|------------------------|
| 2023 – 2027 | <b>B.Tech, Computer Science (Data Science Specialization)</b><br>SRM Institute of Science and Technology (SRMIST) | <b>CGPA: 8.19 / 10</b> |
| 2023 / 2021 | Class 12th CBSE: 64.6%   ·   Class 10th CBSE: 71.8%                                                               |                        |

CERTIFICATIONS

Natural Language Processing – NPTEL

- Completed an NPTEL-certified course covering text preprocessing, tokenization, stemming, lemmatization, language modelling, text classification, topic modelling, word representations, and NLP evaluation techniques.

SKILLS

|                 |                                                                                                                                                                      |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages       | Python, C, C++, Java, JavaScript, TypeScript, Kotlin                                                                                                                 |
| Frontend        | React.js, TypeScript, JavaScript, HTML, CSS, and Tailwind CSS                                                                                                        |
| Backend         | Node.js, REST APIs, JWT Authentication, Role-Based Access Control, and Batch Processing                                                                              |
| Databases       | SQL, PostgreSQL, MongoDB, Relational Database Design, Queries, Joins, Indexing, and Aggregation                                                                      |
| Core CS         | Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, and Computer Networks                                                          |
| Tools           | Git, GitHub, VS Code, Linux, Ubuntu Server, Docker, Codex, Claude Code, Opencode, CasaOS, Tailscale, and Netlify                                                     |
| AI and ML       | Machine Learning, Supervised Learning, Classification, Regression, Feature Engineering, Data Preprocessing, Model Evaluation, and Hyperparameter Tuning              |
| NLP             | Tokenization, Stemming, Lemmatization, Bag of Words, TF-IDF, Text Classification, Sentiment Analysis, Topic Modelling, Word Embeddings, and Transformer Fundamentals |
| Computer Vision | OpenCV, Image Preprocessing, Thresholding, Edge Detection, Contour Detection, and Image Classification Fundamentals                                                  |
| Libraries       | Pandas, NumPy, Scikit-learn, OpenCV, and Matplotlib                                                                                                                  |
| DSA Practice    | Arrays, Strings, Linked Lists, Stacks, Queues, Trees, Sorting, Searching, Recursion, and regular LeetCode problem solving                                            |